Reg. No. | | | | | | | | | | | [14CS15/25, 13CS15/25]

FIRST/SECOND SEMESTER B.TECH EXAMINATION, DECEMBER 2015

PROBLEM SOLVING THROUGH PROGRAMMING

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any FIVE full questions selecting atleast two from each part.

PART - A

1. a. Discuss about working principle of any two primary storage devices.

   b. List the different kinds of access methods. Explain. **(5+9)**

2. a. Write a program to compute the difference and product of two numbers using functions.

   b. Write a program to count the number of words entered. **(6+8)**

3. a. Discuss about Type casting and give an example program which demonstrates type casting.

   b. Write the trace matrix for the following code:

```
int     main()
{
        int i, n, fib1=0, fib2=1, fib3;
        scanf("%d",&n);
        if(n==1)printf("%d",fib1);
          else if(n==2)printf("%d\n%d",fib1, fib2);
                else
                {
                        printf("%d\n%d",fib1, fib2);
                        for(i=3;i<=n;i++)
                        {
                                fib3=fib1+fib2;
                                fib1=fib2;
                                fib2=fib3;
                                printf("\n%d",fib3);
                        }
                }
                        return 0;
}
```

(6+8)

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4. a. Write the output for the following code and explain:

```
for(i=1; i<20; i++)
{
        if(i%2==0) continue;
        printf("%d",i);
}
```

b. Write the output for the following code and explain:

```
        i=1;
Label1: if(i%4==0)printf("%d",i);
        if(i<=40)
        {
                i++;
                goto Label1;

        }
```

c. Differentiate between **break** and **continue** with example code.

(4+4+6)

PART – B

5. a. Differentiate between automatic and register variables in C with an example program.

b. What is recursion? What happens when there is no terminating condition in a recursive program?

c. Differentiate between extern and static variables in C with an example program.

(5+4+5)

6. a. Write a program to add two numbers using pointers.

b. Write a program to read a string "Jain University" and print it character by character using pointers.

(7+7)

7. a. What are command line parameters? Explain its usage with an example program.

b. Write a program to demonstrate the usage of Pointer Arrays.

(8+6)

8. a. What are self-referential structures? Explain with an example.

b. Define a structure to store Employee details (empno, name, address, salary). Input the details of N employees into the array of structures and display the details of all employees.

(6+8)